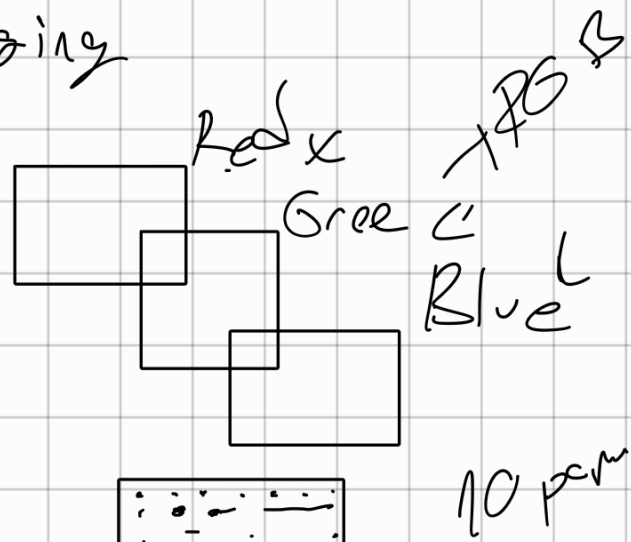
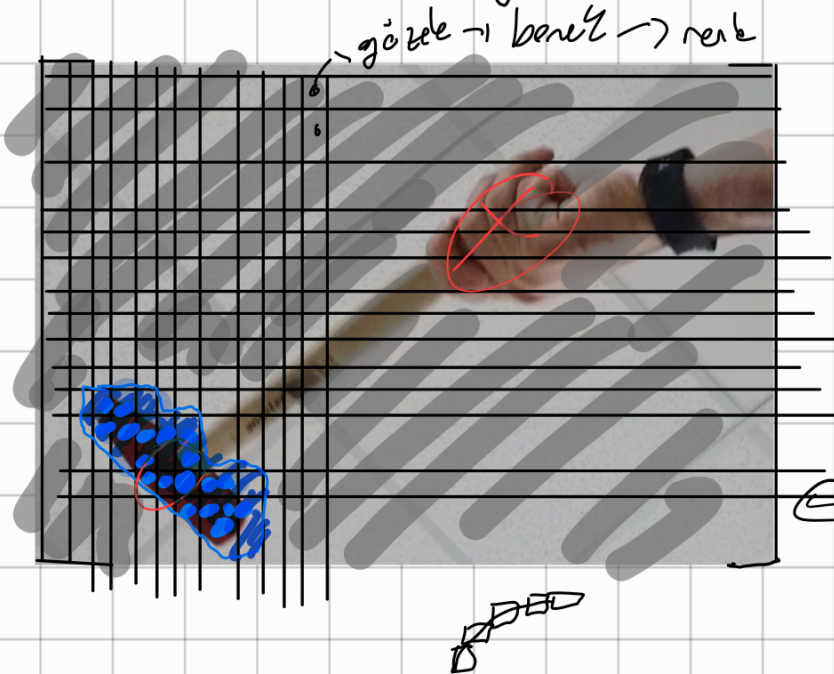


# Camera Detecting and Ranging



Camera sensorü

[...][...][...][...]  
[...][...]

Raw

[...][...][...][...]  
[...][...]

Mask

1 -> White  
0 -> Black

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 |

$$O(n) = n^2$$

Processed - data  
output

[...][...]  
[125, 10, 50, -5, 20...][110, 45...]

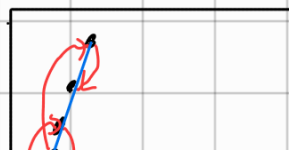
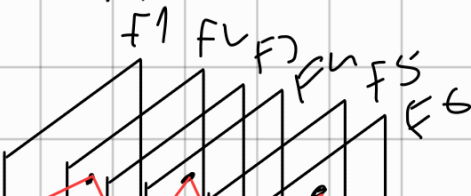
15 FPS

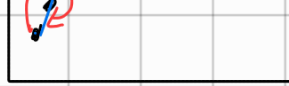
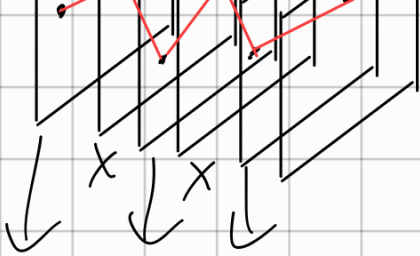
10x

10x

$$\frac{1}{20} \rightarrow 0,05 \text{ sn}$$

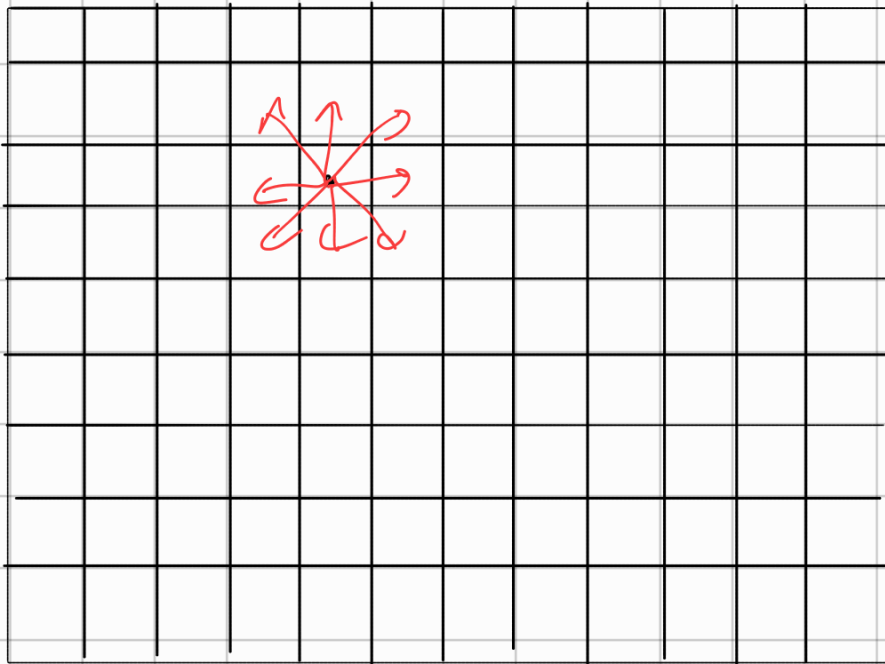
FPS -> Frame Per Second





object tracking

Resolution = Gözün boyutu



9

12

Res = 300 x 300

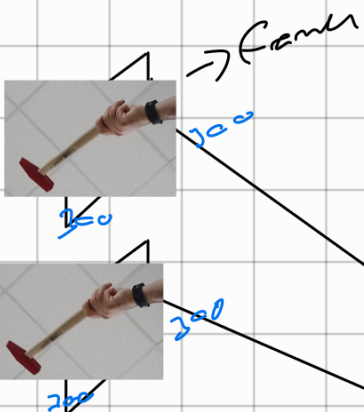
Res = 12 x 9 = 108 piksel

640 x 480 = 307,000 piksel / 30 FPS → 30 x 10ms = 300ms

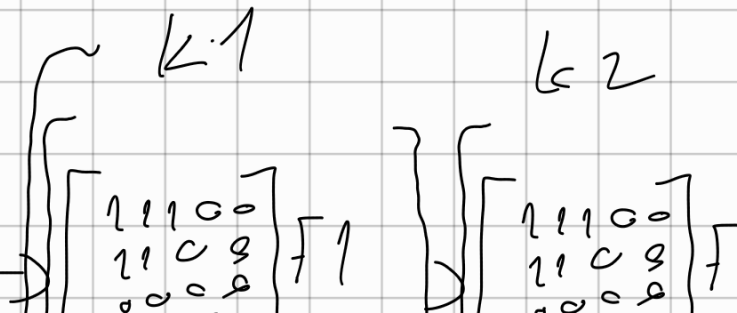
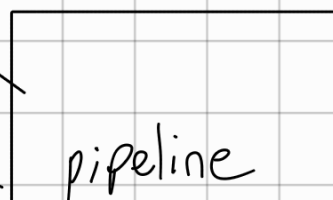
1280 x 720 = 921,600 piksel / 60 FPS → 60 x 10ms = 600ms

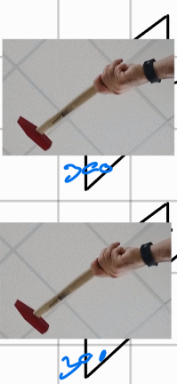
1920 x 1080 = 2,073,600 piksel

0.5 FPS



30sn 30x30

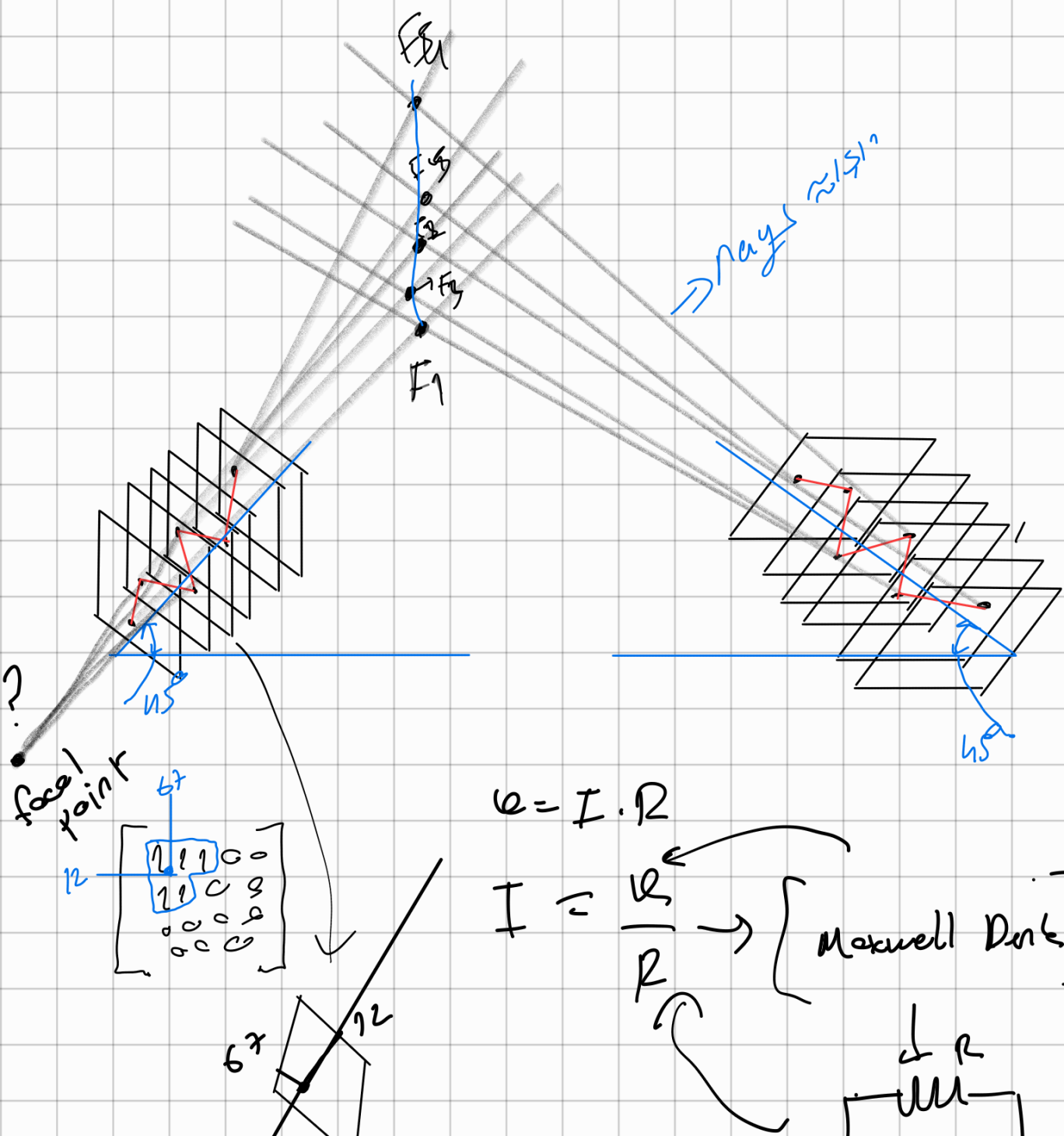




(200x200)

mask.json

$$\begin{bmatrix} \dots \\ \begin{bmatrix} 001110 \\ 001100 \\ \vdots \end{bmatrix} f_2 \\ \begin{bmatrix} 000000 \\ 001111 \\ 001100 \end{bmatrix} f_3 \\ \vdots \end{bmatrix} f_{900}$$



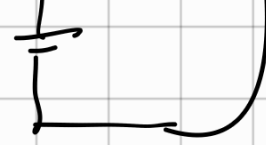
$$\begin{bmatrix} 11100 \\ 11000 \\ 00000 \\ 00000 \end{bmatrix}$$

$$v = I \cdot R$$

$$I = \frac{v}{R} \rightarrow \text{Maxwell Dark}$$

$$\downarrow R \\ \text{mm}$$

face/pe'



İşlem yükü  $\propto$  Genişlik  $\times$  Yükseklik  $\times$  FPS

Örnek:

| Çözünürlük     | FPS | Göreceli işlem yükü | 📄 |
|----------------|-----|---------------------|---|
| 640×480        | 30  | 1× (baz alınan)     |   |
| 1280×720       | 30  | ~3×                 |   |
| 1280×720       | 60  | ~6×                 |   |
| 1920×1080      | 60  | ~13×                |   |
| 3840×2160 (4K) | 60  | ~54×                |   |